

EXCEL FOR DATA ANALYSIS

Assignment - Week 1 and Week 2

Student Name		Student ID	
Class/Section		Submission Date	

Assignment Instructions

- Use the provided Excel workbook and work from a copy of the raw dataset.
- Do not overwrite the original Raw_Student_Data sheet. Keep the original data unchanged.
- Create separate sheets for cleaned data, analysis, summary results, and lookup work where appropriate.
- Use Excel formulas and functions. Do not manually type calculated answers where a formula is required.
- When cleaning data, investigate records before deleting or replacing values.
- Save the completed workbook using the format Fullname_Excel_Assignment.xlsx.
- Submit the completed Excel workbook as instructed by the lecturer.

Assessment Structure

Section	Topic
A	Workbook and Worksheet Formatting
B	Sorting and Filtering
C	Formulas and Cell References
D	Basic Functions
E	Text Functions
F	Date Functions
G	Logical Functions
H	Conditional Functions
I	Data Quality Inspection
J	Remove Duplicates and Blank Rows
K	Missing and Inconsistent Data
	TOTAL

Section A - Workbook and Worksheet Formatting

1. Create a working copy of Raw_Student_Data and rename it Cleaned_Data.
2. Format the header row using bold text and a suitable cell style.
3. Adjust column widths so all headings and values are visible.
4. Apply appropriate number formats to **dates**, **GPA**, **attendance**, **percentages**, and monetary fields where relevant.
5. Use **Freeze Panes** so the heading row remains visible while scrolling.
6. Demonstrate **Hide** and **Unhide on** one selected column.

Section B - Sorting and Filtering

1. Sort student names from A to Z.
2. Sort GPA from highest to lowest.
3. Create a multi-level sort: first by Department, then by GPA from highest to lowest.
4. Use Filter to display only Computer Science students.
5. Use Filter to display only Female students.
6. Use a numeric filter to display students with attendance below 70 or GPA above 3.0.

Section C - Excel Formulas and Cell References

1. Create a calculated column that adds the available student assessment components to produce a Total Score.

Example: =H2+I2+J2 or =SUM(H2:J2)

2. Copy the formula down using relative cell references.
3. Create one calculation that uses an absolute reference to a fixed value stored in a separate cell, such as a **pass mark, fee rate, or percentage**.
4. Example: =K2*\$PS1
5. Create one formula using parentheses to demonstrate the correct order of operations.
6. Add a short note in the workbook explaining the difference between a relative reference and an absolute reference.

Section D - Basic Functions

1. Use SUM - Calculate an appropriate total.
2. Use AVERAGE - Calculate the average of a numeric field such as GPA or score.
3. Use MIN - Find the lowest value.
4. Use MAX - Find the highest value.
5. Use COUNT - Count cells containing numeric values.
6. Use COUNTA - Count non-empty records in a suitable column.
7. Place all summary results in a clearly labeled Analysis or Summary section.

Section E - Text Functions

1. Use TRIM - Remove unnecessary spaces from student names.
2. Use PROPER - Standardize student-name capitalization.
3. Use UPPER - Convert a selected text field to uppercase.
4. Use LOWER - Standardize email addresses to lowercase.
5. Use LEFT - Extract characters from the beginning of a Student ID or another suitable field.
6. Use RIGHT - Extract characters from the end of a Student ID or another suitable field.
7. Use MID - Extract characters from the middle of a text value.
8. Use CONCAT - Combine two or more text values into a new field.

Section F - Date Functions

1. Use TODAY - Display the current date.
2. Use NOW - Display the current date and time.
3. Use YEAR - Extract the year from a date field.
4. Use MONTH - Extract the month from a date field.
5. Use DAY - Extract the day from a date field.
6. Use DATE - Create a date using year, month, and day values.
7. Use NETWORKDAYS - Calculate working days between two appropriate dates.
8. Ensure all calculated dates are formatted correctly.

Section G - Logical Functions

1. Use IF to classify students into two outcomes such as Pass/Fail based on a stated rule.
 - a. If the student totalscore marks above 50 will pass other outcomes fail.
2. Example: `=IF(TotalScore>=50,"Pass","Fail")`
3. Use Nested IF to assign multiple grade categories such as A, B, C, D, and F.
4. Repeat the grade classification using IFS.
5. Use AND inside IF to test two conditions that must both be true, such as GPA and Attendance requirements.
 - a. Example: `=IF(AND(GPA>=3.5,Attendance>=80),"Eligible","Not Eligible")`
6. Use OR inside IF to identify a warning condition when at least one risk condition is true.
 - a. Example: `=IF(OR(GPA<2,Attendance<60),"Warning","Good Standing")`
7. Copy all logical formulas down the dataset and verify a sample of results manually.

Section H - Conditional Functions

1. Use COUNTIF to count students in one selected department.
2. Use COUNTIF to count students in one selected gender or status category.
3. Use SUMIF to calculate the total of a suitable numeric field for one department or category.
4. Use AVERAGEIF to calculate the average GPA, attendance, or score for one department or category.
5. Where possible, place the criterion in a cell and reference that cell in the formula instead of typing the criterion directly.

Section I - Data Quality Inspection

1. Inspect the raw dataset before cleaning it.
2. Identify examples of duplicate records, blank rows, missing values, extra spaces, inconsistent capitalization, inconsistent categories, invalid GPA or attendance values, and mixed/incorrect formats where present.
3. Do not change the Raw_Student_Data sheet during this inspection.

Section J - Remove Duplicates and Blank Rows

1. Record the number of rows before cleaning.
2. Use Conditional Formatting or another suitable method to identify duplicate Student IDs.
3. Investigate the duplicate records before deleting them.
4. Use Data > Remove Duplicates to remove only confirmed duplicates.
5. Identify and delete completely blank rows. Do not delete a valid student row just because one field is missing.
6. Record the final row count and the number of duplicates/blank rows removed.

Section K - Missing and Inconsistent

1. Use Filter to identify missing values in at least three important columns.
2. Use COUNTBLANK to count missing values in at least two columns.

Example: =COUNTBLANK(F2:F301)

3. Decide how each type of missing value should be handled: recover the value, leave blank, use an appropriate label such as Unknown, or exclude it when justified.
4. Do not automatically replace all missing numeric values with zero.
5. Standardize inconsistent values such as Male/male/M, city names, department names, status values, names, and email addresses using appropriate Excel tools or functions.

Required Workbook Structure

- Raw_Student_Data - original dataset, unchanged
- Cleaned_Data - cleaned version of the dataset
- Analysis - formulas, functions, and logical calculations
- Summary - key aggregate results

Note: The assignment is designed to assess both Excel technique and data-analysis judgment. Students should be able to explain why they chose a particular cleaning or formula approach.